

VPNs

At the end of this episode, I will be able to:

1. Describe a VPN
2. Contrast a client to site VPN with a site to site VPN

Learner Objective: *Learn about VPNs and the various types and technologies at use*

Description: In this episode, you will learn about the client to site and the site to site VPN. You will also learn about clientless VPNs and split tunneling.

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- A VPN, or Virtual Private Network, is a technology that creates a secure and encrypted connection over the internet, allowing users to browse the web privately and securely while masking their true location and IP address. It achieves this by routing the user's internet traffic through a remote server operated by the VPN provider, effectively hiding their online activities from potential eavesdroppers and hackers.
 - A client-to-site VPN, also known as a remote access VPN, enables individual users to securely connect to a private network from a remote location, such as their homes or public Wi-Fi networks. In this setup, a client application on the user's device establishes an encrypted tunnel to a VPN server hosted by the organization. This server then authenticates the user and provides them with access to the internal resources of the private network, ensuring that the data transmitted between the user and the network remains confidential and protected from unauthorized access. Client-to-site VPNs are commonly used to facilitate secure remote access for employees or partners requiring access to sensitive company information.
 - A clientless VPN allows users to securely access a private network from a remote location without the need for installing dedicated software on their devices. Instead of relying on a separate client application, users can access the VPN through a web browser, using a web portal provided by the VPN server. Once authenticated, the user gains secure access to internal network resources. Clientless VPNs are particularly convenient as they eliminate the need for pre-installed software on user devices, making them suitable for scenarios where installing a VPN client may not be feasible or practical, such as on public computers or devices with limited administrative privileges.
 - Split tunneling is a networking configuration in which a user's internet traffic is divided or "split" between two separate paths, allowing some data to pass through the encrypted VPN tunnel while other data accesses the internet directly.
 - A site-to-site VPN establishes a secure and encrypted connection between two separate physical locations or networks over the internet, enabling seamless communication between them. In this configuration, dedicated VPN hardware or software on routers or firewalls at each site creates a virtual tunnel through which data can securely traverse. Site-to-site VPNs are commonly used by businesses to connect branch offices, allowing them to share resources and communicate confidentially over the public internet as if they were part of the same private network.
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Additional Resources:

What is a VPN

<https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-vpn>